

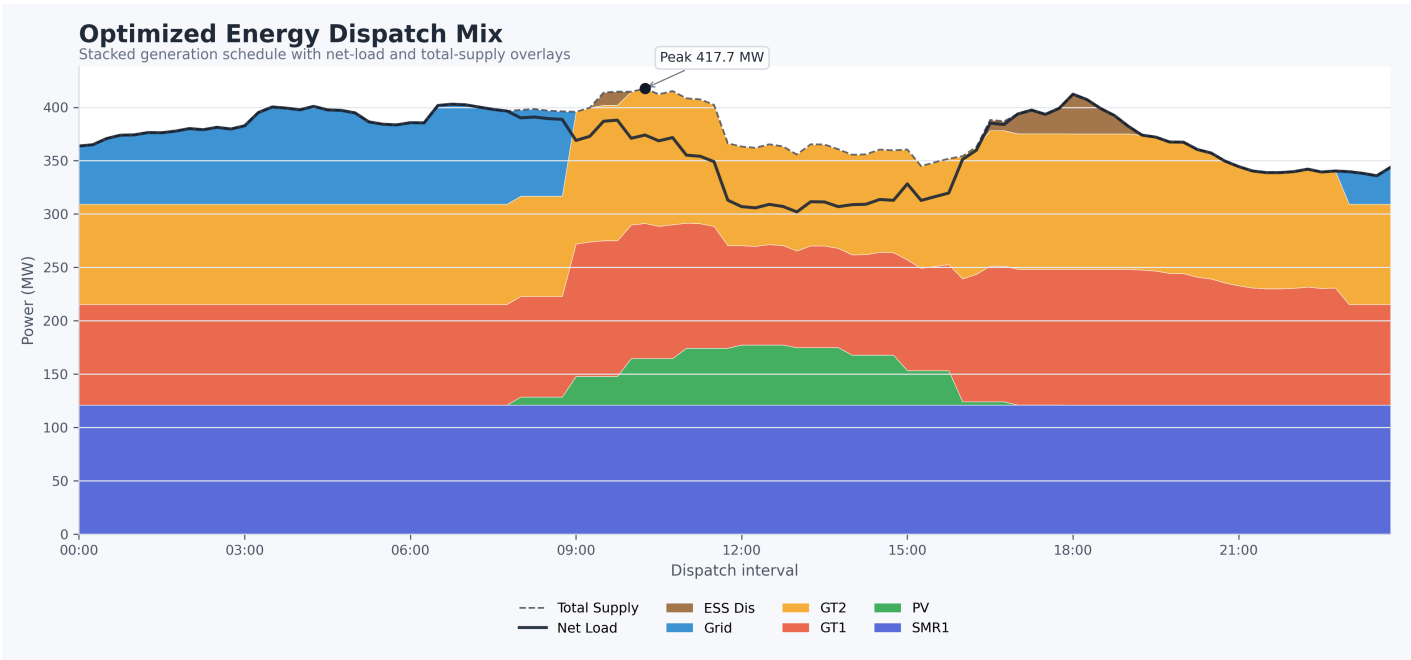
Data Center Energy Dispatch Report

Scenario-based optimization | PV · GT · SMR · ESS microgrid

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TOTAL COST	PV RENEWABLE SHARE	GRID IMPORT	GENERATOR OUTPUT	PEAK DEMAND
1,007,403,126	3.6	2,934.7	31,813.5	417.7
KRW	%	MWh	MWh	MW

ENERGY DISPATCH MIX



AI ANALYSIS REPORT

TECHNICAL REPORT ON MICROGRID ENERGY SYSTEM

EXECUTIVE SUMMARY

This report provides an in-depth analysis of a microgrid energy system comprising photovoltaic (PV) panels, a gas turbine (GT), a small modular reactor (SMR), and an energy storage system (ESS). The analysis focuses on system configuration, cost structure, dispatch strategy, and time-of-use (TOU) operation strategy. The report concludes with an overall assessment and specific recommendations for optimizing the system's performance. Data limitations and assumptions are also discussed.

SYSTEM CONFIGURATION

The microgrid system is configured with a total installed capacity of 557.3 MW, distributed among four key components: PV (56.3 MW), GT (340.0 MW), SMR (121.0 MW), and ESS (40.0 MW with a storage capacity of 160.0 MWh). The SMR operates as a baseload generator, maintaining a constant output of 121.0 MW, which represents 21.7% of the total installed capacity. The GT serves as a flexible generation source with an average output of 210.4 MW, accounting for 37.8% of the total capacity. The PV system contributes to peak renewable support with an average output of 13.5 MW, while the ESS is utilized for peak shaving and arbitrage with an average discharge of 2.5 MW.

COST STRUCTURE

The cost structure of the microgrid system is characterized by a total cost of 1,007,403,126 KRW, with a variable cost of 899,536,460 KRW, representing 89.3% of the total cost. The baseload generation cost is 23 KRW, and the fixed base cost included in the total cost is 107,866,666 KRW. The cost ratios indicate a significant reliance on variable costs, which may be attributed to the flexible operation of the GT and the integration of renewable energy sources.

DISPATCH STRATEGY

The dispatch strategy for the microgrid system is optimized for each generation source. The SMR operates at full capacity with a 100.0% capacity factor, providing a stable baseload. The GT, with a capacity factor of 61.9%, adjusts its output to meet demand fluctuations, averaging 210.4 MW. The PV system, with a capacity factor of 24.0%, contributes 323.8 MWh of energy, primarily during peak periods. The ESS, with a discharge-based capacity factor of 6.3%, supports peak shaving and energy arbitrage, discharging a total of 60.8 MWh.

TOU OPERATION STRATEGY

The TOU operation strategy is divided into off-peak, mid-peak, and on-peak periods. During off-peak, the SMR maintains its output at 121.0 MW, while the GT operates at 188.3 MW, and the PV contributes minimally at 0.8 MW. The ESS does not discharge during this period but stores 73.6 MWh. In the mid-peak period, the GT output increases to 241.2 MW, and the PV output rises to 3.9 MW, with the ESS discharging 8.0 MW and storing 31.9 MWh. During on-peak periods, the PV output peaks at 41.3 MW, the GT operates at 211.1 MW, and the ESS discharges 0.7 MW while storing 73.3 MWh.

RECOMMENDATIONS

The microgrid system demonstrates a balanced integration of renewable and conventional energy sources, with effective utilization of the ESS for peak shaving and arbitrage. However, the following recommendations are proposed to enhance system performance:

1. Increase PV Utilization: Explore opportunities to increase the capacity factor of the PV system by optimizing panel orientation and incorporating advanced forecasting techniques to better align generation with demand.
2. Enhance ESS Efficiency: Investigate the potential for increasing the discharge capacity factor of the ESS by implementing advanced energy management systems and exploring additional revenue streams through ancillary services.
3. Optimize GT Operation: Consider strategies to further optimize the operation of the GT, such as implementing demand response programs or integrating real-time market data to adjust output more dynamically.

LIMITATIONS & ASSUMPTIONS

The analysis is based on a 24-hour horizon, and the data provided does not include specific operational constraints or environmental conditions that may affect system performance. It is assumed that the SMR operates continuously at full capacity and that the GT can adjust its output as needed. The PV generation is assumed to be subject to typical diurnal variations, and the ESS is assumed to have no degradation or efficiency losses. These assumptions may limit the applicability of the findings to real-world scenarios where such factors could influence system operation.